learning can be done in low-end machine, and it requires less time for training, but it consumes more time for testing. But, in case of deep learning, implementation can be done in high-end machine and consumes less time for testing, but requires longer time for training.

3 Deep Learning CNN Model

Convolution neural networks (CNNs) [2] are the most popular deep learning architecture. Mainly, CNN is used in image-related problems [11]. CNN can be used in Language Modeling, Sentiment Analysis, Language Translation, and more. The use of CNN is widely increased because of its capability of extracting important patterns from the image without the involvement of human beings. For example, for a given image set of dog and cat, CNN trained a model having a large number of datasets, and the model automatically identifies that the particular image is a dog or cat by having some important features. CNN model is a very efficient and pure model, and it has many layers of convolution, pooling and fully connected layers. CNN [3, 4] shares the weight parameter in fully connected (FC) layers and so reduces the time overhead in the calculation of every weight.

3.1 History Behind CNN

Two neurophysiologists—David Hubel and Torsten Wiesel—in 1959. In their publication, entitled "Receptive fields of single neurons in the cat's striate cortex", described core response properties of visual cortical neurons as well how a cat's visual experience shapes its cortical architecture. The duo ran some pretty elaborate experiments. They placed electrodes into the primary visual cortex area of an anesthetized cat's brain and observed, or at least tried to, the neuronal activity in that region while showing the animal various images (Fig. 1). A few months into the research, they noticed, rather accidentally, that one neuron fired as they were slipping a new slide into the projector. This was one lucky accident! After some initial confusion, Hubel and Wiesel realized that what got the neuron excited was the movement of the line created by the shadow of the sharp edge of the glass slide.

The researchers discovered some neurons present in the primary visual cortex as a combination of simple and complex neurons, and they are activated by visualizing some oriented edges [1]. This feature is used as the primary concept in deep learning. For this reason, the first digital image scanner was invented. In 1959, an apparatus was developed by Russell Kirsch along with his colleagues for transforming images into grids of numbers so that the machine can understand the images.

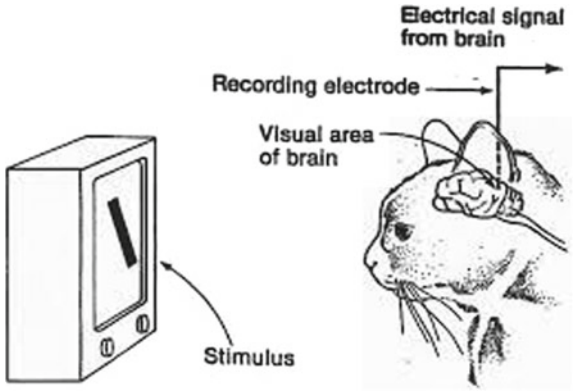


Fig. 1 Experiment on cat's brain

Table 1 ANN versus CNN

ANN	CNN
ANNs take a vector of inputs and produce an output through another hidden layer vector fully connected to that input	But CNN exploits the structure of image through different layers of it which is discussed in the next section
ANN consumes more time in estimating weights. Ex: 224 × 224 × 3 weights need to be calculated for a single neuron in the output layer for a RGB image of size 224 × 224	But in CNN, less time incurred, as it is leading sparse connections between input and output neurons
There is no parameter sharing between output neurons which take place	There is a parameter sharing between output neurons of CNN

3.2 Reason Behind Choosing of CNN Over ANN

Nowadays, CNN is chosen for every application. The CNN has some architectural advantages over ANN, which are discussed in given Table 1.

3.3 CNN Architecture

CNN takes an image in Fig. 2 in three dimensions (width, height, depth). So, neurons are arranged in that manner. Each layer of CNN takes 3D input and transforms it into the 3D output of neuron activations. If it is an RGB image, then depth is 3, and height and width will be the dimensions of the image. CNN works well on image. CNN consists of convolution, pooling layers and fully connected networks. Each layer of CNN is discussed in the next subsection.

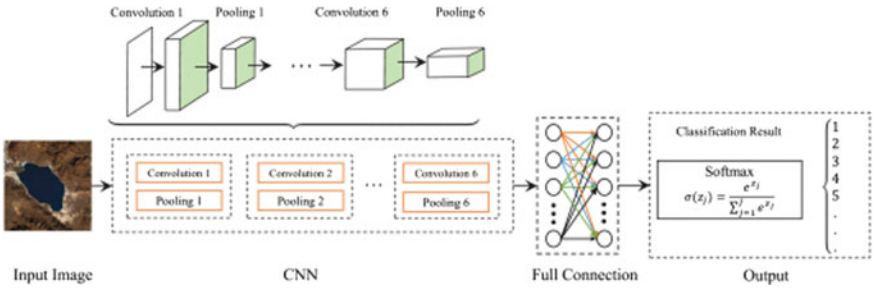


Fig. 2 Image shows the architectural overview of CNN

3.3.1 Convolution

Convolution layer is the vital building block of CNN. It acts as a convolution of two inputs to produce an output. This is the first layer that is applied to input data to get a feature map. It takes input data and a kernel. If the input is an image, then the kernel will be a 2D array consisting of 0 and 1 s. The corresponding output will come out according to the shape of the kernel. The kernel/filter moves to the right of the input image with a certain Stride Value until it parses the complete width. Moving on, it hops down to the beginning (left) of the image with the same Stride Value and repeats the process until the entire image is traversed. Then this would be the input for the next stage.

The convolution conv($I \cdot k$) is a dot product of the input image, I, and the convolution kernel, k. The output is a convolved feature map, F_c. Formally, F_c is obtained by

$$\begin{aligned} F_c &= \text{conv}(i, j) \\ &= (I \otimes K)(i, j) \\ &= \sum m \sum n \; I(m, n)k(i - m, j - n) \end{aligned} \tag{1}$$

$\otimes$ denotes a 2D discrete convolution operator in Eq. 1, which shows that the convolution kernel "k" spatially slides over the input image I to compute the element-wise multiplication and sum to produce an output a convolved feature map F_c. For example, let us consider the convolution operation of input image 5×5 with the 3×3 kernel to produce a 3×3 convoluted feature map. For that, the multiplication of each element of input data with each element of the kernel is done and the final output of each multiplication operation is stored in the feature map of 3×3. The size of the feature map depends on the size of the kernel you have taken. If the image is an RGB-colored image, then the multiplication operation is carried out 3 times as a multiplication of red image with the kernel, multiplication of blue image with kernel and multiplication of green image with the kernel. Finally, all three outputs are combined to form convolution output.

3.3.2 NonLinearity

Any neural network becomes powerful when it has some nonlinearity in the input. Each output of the convolution layer is gone through the activation function. ReLU activation units are used in the output of the convolution layer and stores the output ReLU operation in the feature mapped array (F_T) in Eq. 2.

$$F_T = \text{ReLU}(x_i) = \max(0, x_i) \tag{2}$$

3.3.3 Stride and Padding

How the convolution filter moves on the input image is decided by stride which specifies how many numbers of locations the filter slides on the image. The size of the convolution filter must be less than the input image size. If it is not matched, then padding of zero values is added in the image to make a match with the size of the convolution filter. Generally, the padding on CNN is useful as it is used to preserve the size of the feature map. Every layer of CNN the input is shrunk, so padding is used to get back the desired dimension. The stride of 1 moves the filter one pixel at a time, and similarly, the stride of 2 moves the filter two pixels at a time. Each convolution layer takes input volume in Eq. 3 to produce output volume whose size is decided by three hyperparameters such as depth (no. of filters used), stride and zero padding. How many neurons (say N) will be fitted in the output layer of the convolution layer can be computed as the formula.

$$N = (I_W - C_F + 2Z_P)/(S_z + 1) \tag{3}$$

where I_W = Size of input volume, C_F = Size of the filter, Z_p = Amount of zero padding, S_z = Applied stride.

3.3.4 Pooling

Pooling is the next layer of the convolution layer, which is used to reduce the dimension. It is useful for reducing the number of parameters that yield less training time. Padding is used to combat overfitting. It is used to downsample each feature map with height and width, but depth remains the same. There many types of pooling available; few of them are max pooling and average pooling which are the most commonly used. Max pooling yields the maximum value among all values in the pooling window. The average pooling (F_p) can be computed as in Eq. 4.

$$\text{Average pooling}\left(F_p\right) = \text{pool}(i, j) = \frac{1}{M} \sum mXi, j \tag{4}$$

3.3.5 Fully Connected (FC)

The fully connected layer is the last layer of CNN architecture. This is the same as ANN architecture. This Fc layer takes input from other layers and transforms them into specific no. of classes which are already decided by the network. The output layer of the FC layer is computed for error calculation. Then, a loss function (SVM/Softmax) is defined to compute the gradient of error. These errors are propagated backwardly to update weights and bias in backpropagation neural network. In one forward and backward pass, one cycle is completed for training.

4 CNN Application

Because of CNN popularity, it has many applications and used in extensive manner. Some of them are discussed below.

CNN can be trained by taking some medical images to identify particular disease. Kido et al. in 2018 [2] used an image-based CADX with help of CNN, which was used to differentiate different kinds of lung abnormalities such as nodules and diffuse lung diseases with a mean accuracy of 95.2% without data augmentation and 99.4% with data augmentation. Lu et al. in 2019 proposed an effective framework for script identification by integrating local CNN and global CNN, both of which are based on ResNet-20 [4]. They implemented on CVSI-2015 and ICDAR-2017 datasets which showed the highest overall accuracy than on SIW-13 and MLe2e datasets. CNN can be used in the multimedia area to recognize human activity from video frames [5]. Wang et al. in 2016 used some frames from the video and extract sequence features, but using CNN, the output is fed into long short-term memory (LSTM) model for video activity recognition. Jiang et al. in 2019 used two CNN algorithms parallelly [6] for multi-spectral RGB-NIR image classification. First, RGB-CNN algorithm is used in between R, G, B bands of aerial image for extracting features, and second, NIR-CNN algorithm is used to extract features from R, G, B and NIR bands of that image. Both features are fused in a fully connected layer and then classified. By using this method, RGB-NIR data are fully exploited. Lei et al. in 2019 used dilated CNN [5] instead of taking convoluted kernels for image classification. Secondly, they used hybrid dilated CNN which is a stack of dilated CNN having with different dilation rates, to reduce loss incurred in dilated CNN. As compared to traditional CNN [3, 4], this dilated CNN performs better with reduced training time by 12.99% and with the improvement of training accuracy by 2.86%. Secondly, with the HDC model, it reduces the training time to 2.02% and improves the accuracy of training by 14.15% and also testing by 15.35%. CNN can be used with other algorithm to detect a smile in human face. Qu et al. [7] compared two techniques of face detection algorithm such as face recognition using a color algorithm and AdaBoost Haar-like feature algorithm [8, 9]. After the face recognition, CNN is used for smile detection in that face. They also trained CNN with different optimizers [10]. CNN with RMSprop with momentum shows the highest accuracy of 92.09%, while CNN with Adam

optimizer shows a maximum accuracy of 92.09%. But it shows only 69.53% with its original image. In the whole picture, the location of the face must be detected by using some face detection algorithm. Then CNN is applied to find special features (in reference [11], CNN was applied for a smile detection) depending upon your cause of research.

5 Conclusion

Deep learning provides a vast field, where image processing can be done with many available algorithms. Deep learning has practicability to solve many real-life problems. It has the ability of unsupervised learning with real-world datasets. So, CNN is one of the best algorithms of deep learning which performs better with the highest accuracy. CNN models are widely used by the researcher for many purposes. Some of them are discussed in this review.

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An Optimized Method for Arrhythmia Classification Using Artificial Neural Network



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Abstract Research based on biomedical data analysis is increasing day by day. Cardiac problem is one of major causes of death worldwide. Accurate classification and analysis of cardiac data is more essential to reduce the death rate. In this work, authors have classified cardiac data collected from UCI repository by using optimized model of artificial neural network (ANN). Weights are optimized using modified genetic algorithm (MGA) technique. Five different types of arrhythmia data are considered and classified. The results found about 90% accuracy. The obtained result is compared with some earlier works, and it is found that ANN-MGA is performing better.

Keywords ECG · Arrhythmia · ANN · Optimization · MOGA

1 Introduction

Electrocardiogram (ECG) is one of the important sources for the analysis of any kind of cardiac disease. Electrical impulses that occur during each heartbeat can be visualized in ECG. Accurate analysis of the ECG is most important in the diagnosis of heart disease. Any changes in a normal ECG can occur during various cardiac abnormalities like atrial fibrillation, ventricular tachycardia, myocardial infarction, hypokalemia, etc. Arrhythmia is a type of cardiac illness, and it can be detected by strongly analyzing waves in ECG [1]. It causes due to the abnormal electrical activity in the heart. In this case, the heartbeat rhythms fast or slow. The factors that

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cause cardiac arrhythmia are smoking and alcohol consumption, mental stress, diet, diabetes, etc.

Due to the manual data analysis system, it is difficult to extract useful information from the clinical data. Computer-based automated disease diagnosis system will be most useful in medical sectors. The system will automatically take the decision by analyzing the data collected through a different medical test. Data includes ECG, ultrasonic images, and MRI which are useful for analysis and diagnosis. Machine learning-based medical data analysis is one of the advanced technologies that can reduce human interaction by enhancing the machine efficiency and will be cost-effective. For early detection of cardiac arrhythmia, a real-time automatic ECG analysis system will be the best support to the clinicians [2]. The characteristic of the ECG varies for different patients in different physical and temporal conditions. Due to these variations in ECG, the task is very difficult for analysis. This requires to develop an automatic ECG classification system [3]. For development of accurate ECG classifier, it is important to extract the useful features from data. Here, in the proposed work, an ANN-based classifier is designed with multiple offspring genetic algorithm (MOGA) optimization technique for the classification of five different types of cardiac arrhythmia.

In the course of the most recent thirty years, the genetic algorithm and their mixtures have been applied to different optimization problems, by reason of their various preferences: free subordinate attributes, straightforward readiness of the optimization model, the parallel idea of the pursuit, and so forth. One urgent issue for the genetic algorithm achievement, particularly for the troublesome issues, is to maintain a strategic distance from the untimely intermingling of the calculation to imperfect areas. The premature convergence of a genetic algorithm emerges when the qualities of some high evaluated genes rapidly achieve overwhelm the population, obliging it to merge to a neighborhood optimum. For this situation, the genetic operators cannot create further descendents better that the guardians the calculation capacity to proceed with the quest for better arrangements is in this way considerably diminished. To keep away from the premature convergence, a genetic algorithm is basic to safeguard the population decent variety during the development.

2 Related Literatures

Automated ECG classification can help the cardiologist for the diagnosis of any type of cardiac abnormalities. In the last few decades, several algorithms have been developed by the researchers for the automatic classification of the cardiac signal. Preprocessing, feature extraction, and classification are the three basic steps in ECG signal classification, and multiple methods have been applied by the researchers for each of these processes. Classification of normal and coronary artery disease was performed by applying higher-order statistics and spectra (HOS) method [4]. From each heartbeat, HOS bispectrum and cumulants features were extracted. For dimensionality reduction, authors have taken PCA. KNN and decision tree classifiers